# Replicating the Conservative Formula

Yingni Chang and Yikai Shen · MTH 9897 Project 8 · CRSP replication, 1929–2024

Paper: Blitz and van Vliet (2018), The Conservative Formula, Journal of Portfolio Management 44(7)

Replicate the US results on the paper's sample, 1929–2016

Then examine factor exposure, later periods and trading costs

Main recent comparison: 1990–2024; the paper ends in 2016

# What we test

The rule: 1,000 largest US stocks → low-volatility half → rank on 12-1 momentum and net payout yield → best 100, equal weight, quarterly

1. Known factors: is the return explained by Carhart (primary) or FF5 + momentum (robustness)?
2. Decay: fixed periods 1929–1962, 1963–1989, 1990–2016, 2017–2024, plus rolling 120-month windows
3. Costs: net of 10 and 30 bps per dollar traded; break-even cost of the excess return and the alpha

All estimates with Newey–West (12 lags) or block-bootstrap intervals

# The data

CRSP monthly stock file via WRDS, extracted 2026-10-13; months 1925-12 to 2024-12

3,954,591 security-months, 26,746 securities, 22,869 delistings (2530 imputed, 317 set to zero)

Universe: 393 stocks in December 1928; the full 1,000 from 1964-06

One share class per company; dividends implied from total minus price return

Held stocks without a monthly return: stale-price carry or in-gap price ratio, every case counted

No release identifier on WRDS: the extract is identified by time stamp and file hashes

# The portfolio rule

Paper order: history screen → top 1,000 by market cap → low-volatility half → ranks among complete signals

Weights drift between rebalances; delisted holdings become cash; costs charged per dollar traded

Controls: Speculative (high-volatility half, weakest scores), standalone signals, Small, CRSP VW market

Excluded: Value (no book equity in CRSP) and Table 2 panels C–D (factors not in our sources)

| **deviation** | **our convention** |
|---|---|
| Share-class selection | largest common class of each PERMCO per month (shrcd 10/11) |
| Size measure | security market cap for the top-1,000 cut |
| Dividends | implied from ret - retx (legacy retx includes non-ordinary distributions) |
| Missing-signal rule (NB-007) | partition first; complete cases ranked |
| Small early universe (DATA-001) | all history-eligible names when < 1,000 (393 at 1928-12; 500-name pool from 1964-06) |
| Missing monthly returns (DATA-002/010) | stale carry (0) or in-gap price ratio |

# Building the signals

Volatility: annualized standard deviation of 36 consecutive monthly total returns

Momentum: compound 11 monthly price returns, excluding the most recent month

Dividend yield: trailing 12 months of split-adjusted dividends divided by the current adjusted price

Net payout yield: dividend yield + 1 − latest adjusted shares / average adjusted shares over 24 months

A falling share count raises payout yield; new share issuance lowers it

Quarter-end information determines next month's holdings. No forward return enters selection.

# Our results versus the paper

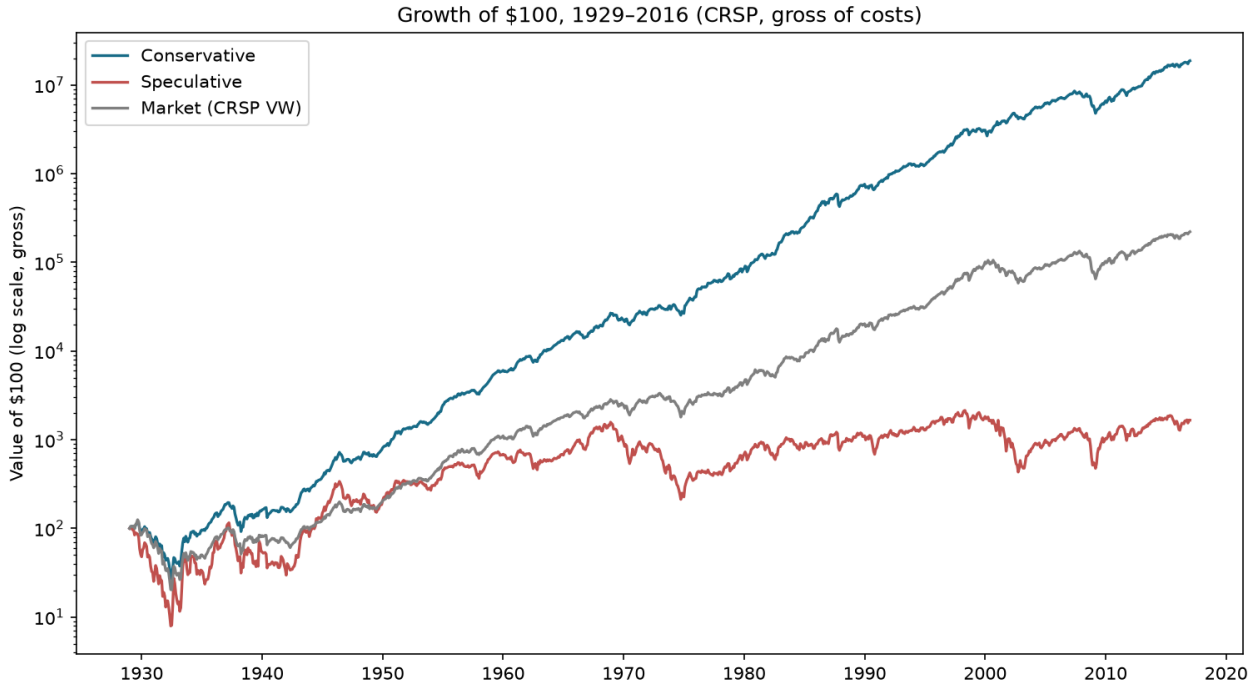
Conservative 14.80% compounded, 17.06% volatility (paper 15.1%, 16.5%)

Market 9.15% (paper 9.3%)

Speculative 3.24% with 35.9% volatility

Conservative beats the market in 9 of 10 decades

Turnover 38.9% a quarter vs the paper's 32% (unexplained; measure not stated)




| paper / ours | Formula | Market | Small | Momentum | Lowvol | NPY |
|---|---|---|---|---|---|---|
| Return (compounded) | 15.1% / 14.8% | 9.3% / 9.1% | 12.0% / 13.0% | 15.9% / 15.5% | 10.2% / 10.4% | 14.5% / 14.2% |
| Volatility | 16.5% / 17.1% | 18.7% / 18.7% | 31.4% / 33.5% | 25.7% / 26.1% | 13.3% / 14.4% | 20.1% / 21.1% |

# Factor-adjusted returns

CMS (Conservative − Speculative), 1963–2016: CAPM 14.20% vs 14.0%; FF5 8.69% vs 8.8%; FF5 + MOM 3.12% vs 3.3%

1929–2016: CAPM 12.33% vs 13.9%; Carhart 7.74% vs 8.8% (full-sample gap remains)

Conservative long-only: Carhart alpha 3.23% (t 4.63)

FF5 + momentum, 1963–2016: −0.11% (t −0.14) — adding profitability and investment reduces alpha

CMS net of 30 bps, both legs paying: Carhart 5.78% (t 3.99)

# How costs change the result

Annual cost estimate: turnover × 8 × cost; quarterly turnover 38.9% vs 32%

Net of 30 bps, 1929–2016: 13.73% compounded

1990–2024 break-even: 52 bps (excess return), 36 bps (alpha)

2017–2024: gross excess return and alpha already ≤ 0, so no break-even

Early-period break-evens are high, but so were real trading costs then

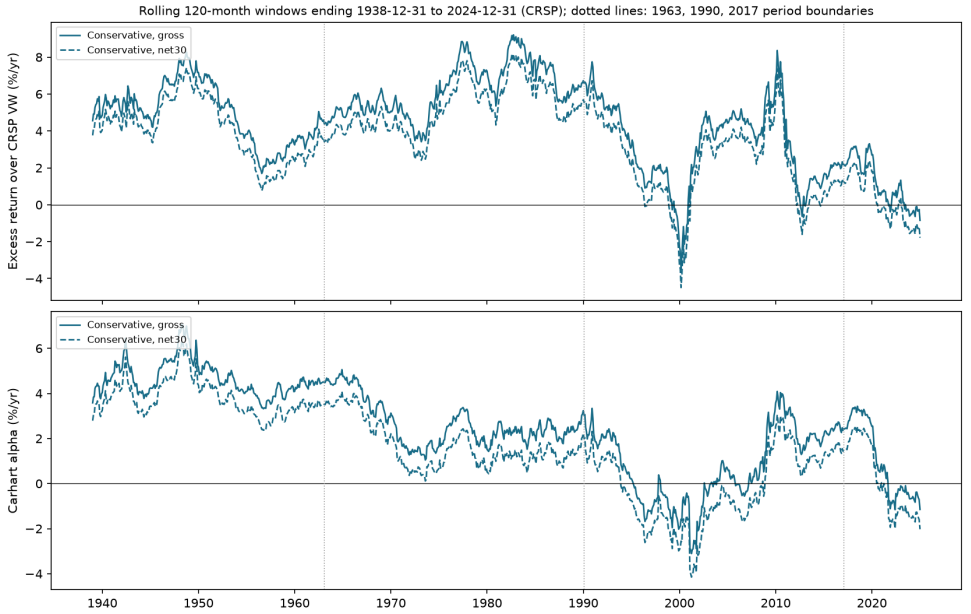
| break-even, bps | alpha | excess return |
|---|---|---|
| 1929–1962 | 162 | 178 |
| 1963–1989 | 66 | 187 |
| 1990–2016 | 52 | 84 |
| 1990–2024 | 36 | 52 |
| 2017–2024 | none | none |

# Results by period

Carhart alpha (gross): 4.7% → 2.1% → 1.7% → −1.2%

Carhart alpha is statistically significant in the two earlier periods

2017–2024 is only 96 months: interval [−5.1%, 2.7%]




| period | months | excess, gross | Carhart α gross (t) | Carhart α net 30 (t) |
|---|---|---|---|---|
| 1929–1962 | 408 | 5.2% | 4.7% (5.03) | 3.8% (4.14) |
| 1963–1989 | 324 | 6.4% | 2.1% (2.90) | 1.2% (1.56) |
| 1990–2016 | 324 | 2.7% | 1.7% (1.34) | 0.7% (0.56) |
| 2017–2024 | 96 | −1.7% | −1.2% (−0.61) | −2.1% (−1.06) |
| 1990–2024 | 420 | 1.7% | 1.2% (1.05) | 0.2% (0.18) |

# The recent sample: 1990–2024

Gross: excess over market 1.69% [−2.0%, 5.4%]; Carhart alpha 1.17% (t 1.05)

Net of 30 bps: excess 0.73% [−3.0%, 4.4%]; alpha 0.20% [−2.0%, 2.4%]

FF5 + momentum alpha, net of 30 bps: −2.00% (t −2.33)

Still lower risk: 12.2% compounded with 13.5% volatility (market 10.1%, 15.2%)

Cost-inclusive CMS alpha 3.72% (t 2.22)

# Dividends and share repurchases

NPY = trailing dividend yield + 1 − latest adjusted shares / 24-month average adjusted shares

Repurchases raise NPY; share issuance lowers it. Splits must leave the adjusted share count unchanged

1929–2016 gross CAGR: 14.80% using NPY versus 13.44% using dividends alone

Paired net-30 arithmetic gain from NPY: 1.22% per year [0.56%, 1.87%]

Same universe, low-volatility pool, momentum signal, weights, quarterly dates, and costs in the comparison

Share-count change measures net repurchases after issuance; it is not reported buyback cash flow

# Industry allocation and stock selection

Paper scores are compared across sectors (page 4); the professor's sector-aware variant is reported separately

1929–2016 gross arithmetic excess return: 4.80% per year

Industry allocation 0.82% + stock selection 3.80% + reference-to-CRSP-market difference 0.18%

Reference: cap-weighted formation universe, quarterly constituents and industries fixed, weights drift, delistings to cash

Sector variant: 100 slots proportional to industry security counts; apply the rule separately within each FF12 industry. Gross CAGR 14.51%, net-30 13.37%

The sector variant is exploratory and changes the paper's global low-volatility screen. Contributions are arithmetic, not causal

# Signal tests

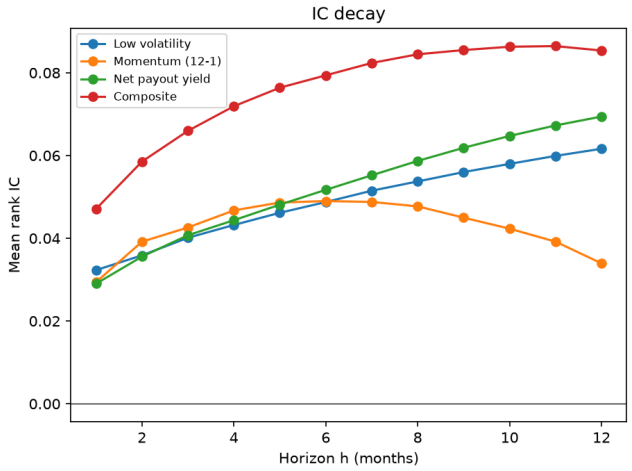
Composite rank IC (1 month): 0.047 (NW t 8.8)

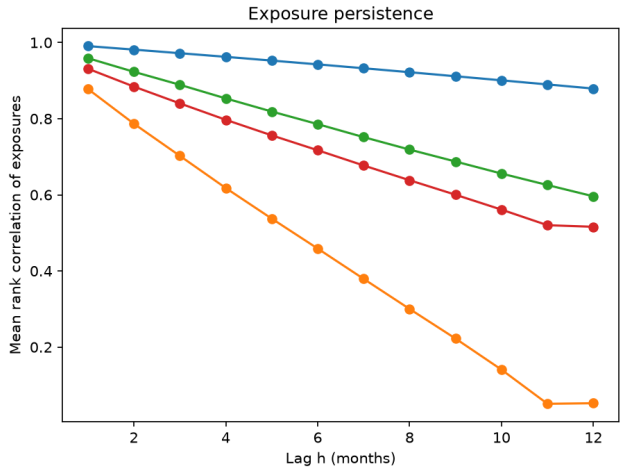
Decile spread D10−D1: 1.02% (1963–1989), 0.68% (1990–2016), 0.40% (2017–2024)

Factor-mimicking paper portfolio: IR 0.59 at K = 1, 0.29 at K = 12

2017–2024 IR at K = 1: 0.29

Low volatility: positive rank IC, flat mean spread (right-skewed high-volatility deciles)



# Long/short and linear optimization

Decile long/short, n = 3, 10 bps, borrow: 0.64% a year, IR 0.03, max drawdown −100.0%

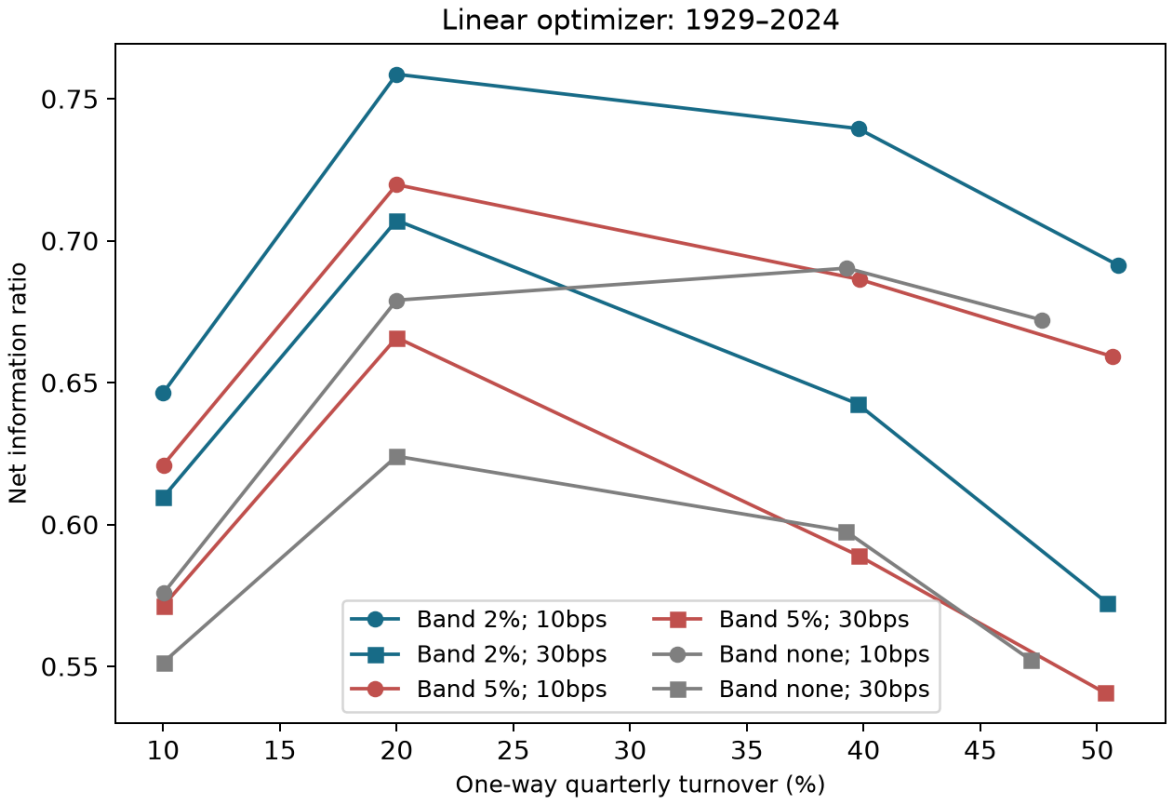
Monthly re-weighted cohorts: chosen after the pre-specified buy-and-hold cohorts were ruined (1932-08, 1933-05)

Capacity 2020s ≈ $7.9B (scaled to 2024 market size; 10% of volume; no impact model)

Linear optimizer, 1990–2024 net 30: Carhart alpha 1.31%–1.95% (2–5% bands, 20–40% turnover budget); two of the four 95% intervals exclude zero

Unadjusted intervals, selected from a 24-run grid; 2017–2024 alphas −2.71% to 0.07% (none of the twelve 95% intervals excludes zero); return-scaled alpha is a later sensitivity

We also test a portfolio covariance penalty



# Mean-variance portfolios

Maximize expected return − half risk aversion × portfolio variance − trading costs

Covariance: trailing 36 monthly returns, 50% diagonal shrinkage; the window ends at formation

Return alpha: completed trailing IC × annual volatility × √(3/12) × composite exposure

Quarterly: 2% stock cap, industry bands ±5pp, two-way turnover budget 40%; scaled costs start at 30bps

Two fixed risk aversions (5 and 20); max constraint violation 4.9e-09; turnover relaxations 0

We chose these settings after examining the baseline; an unseen test is still needed

| 1990–2024 | Net CAGR | Volatility | IR | Market corr. |
|---|---|---|---|---|
| γ = 5 | 11.59% | 12.51% | 0.14 | 0.89 |
| γ = 20 | 10.39% | 11.51% | −0.03 | 0.87 |

# A buffer to reduce trading

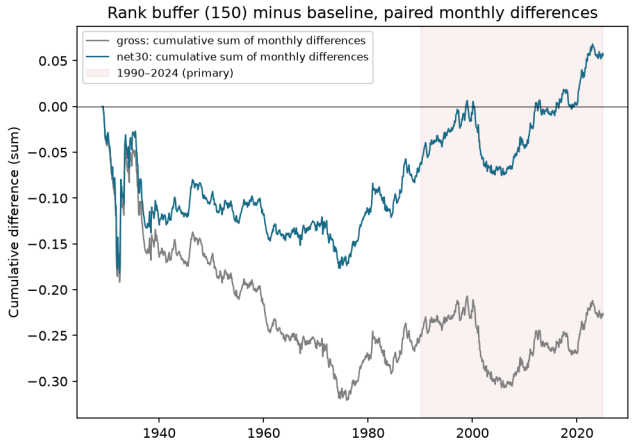
Keep existing holdings while they remain in the eligible pool and within the best 150 ranks

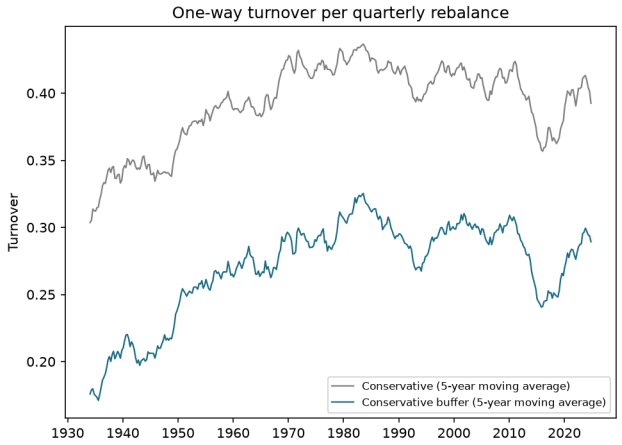
Turnover 39.8% → 28.4% a quarter (28.7% lower)

Net-of-30 paired difference 0.34% a year [−0.10%, 0.78%]: not significant

Gross paired difference 0.06%; the buffer changes holdings as well as costs

The recent period was already seen when we tested this variant



# What we conclude

Return and volatility are close to the paper; turnover and some regression results differ

Much of the historical return is associated with known equity factors

Alphas decay; 1990–2024 net of 30 bps: alpha 0.20% [−2.0%, 2.4%]

The recent net-alpha interval includes zero

Lower volatility is the most consistent feature of the portfolio

Limitations: no CRSP release identifier; implied dividends; share-class and size conventions; turnover above the paper's; selection across grids; a short post-sample

# Running the project

python -m crsp_pipeline.import_web (full web exports; licensed data kept private)

python -m crsp_pipeline build

python -m research.run_all → results/ (aggregates) + results/manifest.json

python build_notebook.py && python execute_notebook.py

python presentation/build_pdf.py (PDF slides and speaker notes)

85 result files with sha256 hashes; data manifest 4816d582000a…